

SDG Goal 3

Good health and well-being

SDG Target 3.9

By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination

SDG Indicator 3.9.1

Mortality rate attributed to household and ambient air pollution

Time series

Attributable deaths due to long-term exposure to fine particulate matter (PM_{2.5}) in the population aged 25 years and older

1. General information on the time series

- Date of national metadata: Mai 4, 2026
- National data: <https://sdg-indicators.de/3-9-1>
- Definition: The time series shows the number of deaths as a result of a disease, which are attributable to fine particulate matter (PM_{2.5}):

Classification is based on ICD-10 codes. ICD-10 is the 10th Revision of the International Statistical Classification of Diseases and Related Health Problems. In its German Modification (ICD-10-GM), it serves as the official classification system for coding diagnoses in outpatient and inpatient care in Germany. The following ICD-10 codes are included:

Type of disease	ICD-10 codes
Chronic obstructive pulmonary disease (COPD)	J40-J44.9; J47-J47.9
Lung cancer	C33-C34.9; D02.1-D02.3; D14.2-D14.3; D38.1
Stroke	G45-G46.8; I60-I62.9; I63-I63.9; I65-I66.9; I67.0-I67.3; I67.5-I67.6; I68.1-I68.2; I69.0-I69.3
Ischemic heart diseases	I20-I25
Diabetes mellitus type 2	E11; Proportions of E13-E14

- Disaggregation: age group, type of disease

2. Comparability with the UN metadata

- Date of UN metadata: December 2023
- UN metadata: <https://unstats.un.org/sdgs/metadata/files/Metadata-03-09-01.pdf>
- The time series is not compliant with the UN metadata, but provides additional information. The reasons for this are:
 - It shows the number of attributable deaths and not, as required by the UN metadata, the mortality rate.
 - It refers to ambient air pollution, specifically urban and rural background pollution, and not, as required by the UN metadata, to household and ambient air pollution.

3. Data description

- The data is derived from the German Environment Agency (UBA). Attributable deaths refer to those deaths that can statistically be attributed to exposure to a specific risk factor.

When calculating the environmental burden of disease, the number of attributable deaths is determined by multiplying the total number of annual deaths from a specific disease – broken down by sex and age group – by the proportion attributable to a specific risk factor (e.g. particulate matter) as the cause of that disease. This proportion is referred to as the Population Attributable Fraction (PAF).

It is assumed that these deaths are attributable to exposure to the relevant risk factor. It follows that, if exposure to this risk factor were completely eliminated, there would be no corresponding disease burden. In other words, if the risk factor were not present, the individuals concerned would have a higher life expectancy.

4. Access to data source

- Health relevance of fine particulate matter (only available in German):
<https://www.umweltbundesamt.de/daten/umwelt-gesundheit/gesundheitsrisiken-durch-feinstaub>
- GBE – Tabelle: Sterbefälle, Sterbeziffern:
https://www.gbe-bund.de/gbe/isgbe.fundstellen?p_uid=gast&p_aid=99629820&p_sprache=E&p_thema_id=3900&p_action=TRT
- Population in 100 meters grid (only available in German):
<https://www.zensus2011.de/DE/Home/Aktuelles/DemografischeGrunddaten.html>

5. Metadata on source data

- Health relevance of fine particulate matter (only available in German):
<https://www.umweltbundesamt.de/daten/umwelt-gesundheit/gesundheitsrisiken-durch-feinstaub>

6. Timeliness and frequency

- Timeliness: May vary.
- Frequency: Annual

7. Calculation method

- Unit of measurement: Attributable deaths
- Calculation: Not applicable.

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SDG Indicator 3.9.1

Mortality rate attributed to household and ambient air pollution

Time series

DALYs due to long-term exposure to fine particulate matter (PM_{2.5}) in the population aged 25 years and older

1. General information on the time series

- Date of national metadata: Mai 4, 2026
- National data: <https://sdg-indicators.de/3-9-1>
- Definition: The time series shows the disease-specific years of healthy life lost attributable to particulate matter.

Classification is based on ICD-10 codes. ICD-10 is the 10th Revision of the International Statistical Classification of Diseases and Related Health Problems. In its German Modification (ICD-10-GM), it serves as the official classification system for coding diagnoses in outpatient and inpatient care in Germany. The following ICD-10 codes are included:

Type of disease	ICD-10 codes
Chronic obstructive pulmonary disease (COPD)	J40-J44.9; J47-J47.9
Lung cancer	C33-C34.9; D02.1-D02.3; D14.2-D14.3; D38.1
Stroke	G45-G46.8; I60-I62.9; I63-I63.9; I65-I66.9; I67.0-I67.3; I67.5-I67.6; I68.1-I68.2; I69.0-I69.3
Ischemic heart diseases	I20-I25
Diabetes mellitus type 2	E11; Proportions of E13-E14

- Disaggregation: age group, type of disease

2. Comparability with the UN metadata

- Date of UN metadata: December 2023
- UN metadata: <https://unstats.un.org/sdgs/metadata/files/Metadata-03-09-01.pdf>
- The time series is not compliant with the UN metadata, but provides additional information.

3. Data description

- The data is derived from the German Environment Agency (UBA). DALYs (Disability-Adjusted Life Years) refer to the healthy years of life lost due to mortality and morbidity that can be statistically attributed to exposure to a risk factor.

To calculate the environment-related disease burden, the total number of disease-specific DALYs – broken down by sex and age group – is multiplied by the Population Attributable Fraction (PAF) for a specific risk factor.

It is assumed that these DALYs are attributable to exposure to the respective risk factor. It follows that a complete reduction in exposure to this risk factor would reduce the corresponding disease burden. In other words: if the risk factor were not present, the life expectancy of the population would increase.

4. Access to data source

- Health relevance of fine particulate matter (only available in German):
<https://www.umweltbundesamt.de/daten/umwelt-gesundheit/gesundheitsrisiken-durch-feinstaub>
- Health in Germany today (only available in German):
https://public.tableau.com/app/profile/robert.koch.institut/viz/Gesundheit_in_Deutschland_aktuell/GEDA_20192020-EHIS
- 12-month prevalence of known chronic obstructive pulmonary disease (COPD) in Germany (only available in German):
<https://edoc.rki.de/bitstream/handle/176904/2783/20vT7kfS6gfmng.pdf?sequence=1&isAllowed=y>
- 12-month prevalence of known diabetes mellitus in Germany (only available in German):
https://edoc.rki.de/bitstream/handle/176904/2580/JoHM_2017_01_gesundheitliche_lage2.pdf?sequence=4&isAllowed=y
- 12-month prevalence of stroke or chronic complaints resulting from stroke in Germany (only available in German):
https://edoc.rki.de/bitstream/handle/176904/2583/JoHM_2017_01_gesundheitliche_lage5.pdf?sequence=4&isAllowed=y

5. Metadata on source data

- Health relevance of fine particulate matter (only available in German):
<https://www.umweltbundesamt.de/daten/umwelt-gesundheit/gesundheitsrisiken-durch-feinstaub>

6. Timeliness and frequency

- Timeliness: May vary.
- Frequency: Annual

7. Calculation method

- Unit of measurement: DALYs
- Calculation:

$$\text{DALYs} = \text{Years of life lost due to death [number]} + \text{Years lived with disability [number]}$$

$$\text{Years of life lost due to death} = D_{a,c} \cdot RLE_a$$

$$\text{Years lived with disability} = P_{a,c} \cdot DW_c$$

With:

$D_{a,c}$ = Age (a) and cause (c) specific number of deaths in the respective years

RLE_a = Age (a) specific remaining life expectancy at age of death

$P_{a,c}$ = Age (a) and cause (c) specific prevalence in the respective years

DW_c = Cause (c) specific disability weight indicating the severity of a disease